ARTICLE IN PRESS

Knowledge-Based Systems xxx (xxxx) xxx

Contents lists available at ScienceDirect

Knowledge-Based Systems

journal homepage: www.elsevier.com/locate/knosys

Non-ferrous metals price forecasting based on variational mode decomposition and LSTM network☆

Yishun Liu, Chunhua Yang, Keke Huang *, Weihua Gui

School of Automation, Central South University, Changsha 410083, China

ARTICLE INFO

Article history:
Received 11 June 2019
Received in revised form 28 August 2019
Accepted 29 August 2019
Available online xxxx

Keywords:
Non-ferrous metals
Price forecasting
Variational mode decomposition
Long short-term memory network
London Metal Exchange

ABSTRACT

Non-ferrous metals are indispensable industrial materials and strategic supports of national economic development. The price forecasting of non-ferrous metals is critical for investors, policymakers, and researchers. Nevertheless, an accurate and robust non-ferrous metals price forecasting is a difficult yet challenging problem due to severe fluctuations and irregular cycles in the metal price evolution. Motivated by the "Divide-and-Conquer" principle, we present a novel hybrid deep learning model, which combines the VMD (variational mode decomposition) method and the LSTM (long short-term memory) network to construct a forecasting model in this paper. Here, the VMD method is firstly employed to disassemble the original price series into several components. The LSTM network is used to forecast for each component. Lastly, the forecasting results of each component are aggregated to formulate an ultimate forecasting output for the original price series. To investigate the forecasting performance of the proposed model, extensive experiments have been executed using the LME (London Metal Exchange) daily future prices of Zinc, Copper and Aluminum, and other six state-of-the-art methods are included for comparison. The experiment results demonstrate that the proposed model has superior performance for non-ferrous metals price forecasting.

© 2019 Elsevier B.V. All rights reserved.

1. Introduction

Non-ferrous metals, which include Zinc, Copper and Aluminum et al. are the basic materials for national economic development, and are the basis of most industries such as mechanical manufacturing, aerospace, electricity, house-hold appliances and so on [1,2]. With the rapid advancement of modern industry, non-ferrous metals play an increasingly crucial role in social development [3,4].

Price forecasting of raw materials in real world is a complex task where different factors are coupled and correlated [5–7]. Non-ferrous metals are the raw materials with both commodity and financial attributes. The prices of them are often influenced by the global economic, supply–demand relationship, production cost, dollar current rate and other factors, so they fluctuate frequently and vary widely. On the other hand, for non-ferrous metals producing and processing enterprises, raw material procurement cost usually accounts for more than 60% of the total cost [8]. The volatile fluctuations and dramatic changes of non-ferrous metals price will directly affect the judgment of manufacturers on the variation tendency of raw material price, and further affect the decision of operation. Therefore, an accurate and robust price forecasting method is of great significance for purchasers to plan and schedule non-ferrous metals.

In the past decades, the forecasting of non-ferrous metals price has attracted wide academic interests, and much effort has been made to forecast the price using various modeling approaches. Generally speaking, these methods can be divided into two categories: single model methods and hybrid model methods. For single model methods, the time series data is modeled and forecasted by an individual model. Dooly and Lenihan adopted lagged forward price model and autoregressive integrated moving average (ARIMA) to predict monthly crash prices of Lead and Zinc. It was found that the forecasting cash prices using ARIMA model led to better results [9]. Aye et al. developed several models and embraced six global factors to predict the return of gold. They found that the dynamic model averaging (DMA) and dynamic model selection (DMS) models outperformed linear models (such as random walk) as well as Bayesian model averaging [10]. Chen et al. proposed a modified gray wave forecasting method which was a graphical forecasting method, to predict the monthly prices of two metals—Aluminum and Nickel with a multi-step-ahead prediction [11].

With the prosperity of machine learning, artificial intelligent methods are gradually applied. Dehghani utilized Bat algorithm

☆ No author associated with this paper has disclosed any potential or pertinent conflicts which may be perceived to have impending conflict with this work. For full disclosure statements refer to https://doi.org/10.1016/j.knosys.2019.105006.

* Corresponding author.
E-mail address: huangkeke@csu.edu.cn (K. Huang).

https://doi.org/10.1016/j.knosys.2019.105006
0950-7051/© 2019 Elsevier B.V. All rights reserved.

to predict the Copper price volatility. The experimental results indicated that the performance of the Bat algorithm was better than the classic estimation methods [12]. Liu et al. and Carrasco et al. utilized decision tree learning algorithm and support vector machine to forecast Copper future price of the London Metal Exchange respectively [13,14]. Malliaris employed artificial neural network to forecast the prices of gold, oil, and the euro [15]. Sivalingam et al. forecasted the gold future price from the historical data of other four commodities like silver, crude oil, S&P 500, and foreign exchange rate, using extreme learning machine [16]. Zhang and Liao examined the forecasting performance of radial basis function (RBF) neural network and hybrid fuzzy clustering algorithm using gold price. The empirical results showed a better performance of hybrid fuzzy clustering algorithm over the former [17]. Lasheras et al. employed ARIMA and two different kinds of ANN models (multilayer perception machine and Elman network) to forecast the Copper spot price from COMEX, and they showed that the performance of both neural network models was superior to ARIMA [18].

Although traditional neural networks have a good prediction ability, the accuracy is still unsatisfactory when facing dynamic and non-linear time-series data. Long short-term memory (LSTM) network [19] has been proved to be one of the most powerful models for processing sequence data like time series. Unlike the traditional ANN, LSTM introduces the memory cells, and the memories of previous inputs can be persisted in the network's internal state. Therefore, LSTM network can explore the inherent abstract features and grasp the hidden-level constant structures in data, then process time series with high prediction ability. With the development of machine learning and deep learning methods, the LSTM model has been widely employed in natural language processing [20], time series prediction [21], emotion analysis [22], and so on. These works indicate that LSTM has excellent ability to learn the dependencies among complex non-linear time series.

For hybrid model methods, the basic idea is to address the shortcomings of single model and generate a synergetic effect in forecasting, which have become the mainstream approaches recently. For example, Zhu et al. used a LHAR-CJ-G model to forecast the Chinese non-ferrous futures (like Copper and Aluminum futures) market volatility in Shanghai Futures Exchange, and effectively improved the explanatory power [23]. Werner and Esteban used a hybrid ANN-GARCH model with regressors to predict the price volatility of gold, silver and copper. Comparison experiments indicated that the incorporation of ANN increased forecasting accuracy compared with traditional GARCH model [24]. Khasher et al. combined ARIMA model with fuzzy logic and ANN to predict the daily price of gold and exchange rate. It overcame the linear and data limitations of ARIMA models, and obtained more accurate results [25]. Li built a gold price forecasting model based on wavelet neural network (WNN) and improved artificial bee colony algorithm [26]. Zakaria et al. employed adaptive neuro-fuzzy inference system (ANFIS) to forecast the volatility of Copper price, and used Genetic algorithm to estimate the parameters of ANFIS model [27]. Diego and Werner developed a framework composed of a set of time series models, non-parametric models and hybrid specifications of both to forecast monthly volatility of Copper market. The results showed that the hybrid adaptive-GARCH-FIS model yielded the best predicting power [28]. Zhang et al. proposed a new HAR-CJN model to predict the Chinese non-ferrous metals futures market, like daily prices of Copper and Aluminum futures. Contrastive experiments indicated that the new HAR-CJN model can describe and predict daily price volatility [29].

The "Decomposition-and-Ensemble" framework, which is motivated by the "Divide-and-Conquer" principle, is one of the most popular hybrid model methods [30]. Here, the goal of decomposition is to disintegrate the original complex time series into several interdependent components. The tendency and fluctuation characteristics of subseries are more evident and help to lighten the burden of forecasting model. On the contrary, the main aim of ensemble is to aggregate the prediction results of subsequences to yield the forecasting consequence of original data [31,32].

For decomposition methods, there are mainly wavelet decomposition [33], Fourier decomposition [34] and empirical mode decomposition (EMD) [35]. Thomas et al. forecasted monthly base metal prices (Aluminum, Copper, Lead and Zinc) by means of an improved wavelet-ARIMA approach [36]. Since wavelet decomposition relies heavily on the mother wavelet functions before decomposing, Fourier decomposition usually generates a large number of meaningless harmonics when dealing with non-linear time series, so they are seldom used in time series processing. However, due to the advantages in decomposing non-linear and non-stationary time series, and the advantages in describing the local time scale instantaneous frequencies, EMD has been used widely [37]. Zhou presented an improved EMD online learning rate-based back-propagation neural network model for gold price forecasting [38]. Xian, He and Lai analyzed the KITCO gold price combining ensemble empirical model decomposition (EEMD) and Independent Component Analysis (ICA) [39]. Wen et al. incorporated complete ensemble empirical mode decomposition (CEEMD) with SVM and ANN for analysis and prediction of gold price [40]. Meanwhile, EMD-based methods generally have the problems of boundary effects, mode overlap, sensitivity to noise and lack of exact mathematical foundations. These limitations may affect the final prediction results. Variational mode decomposition (VMD) [41] is a new multiresolution technique for signal processing, which is entirely non-recursive variation model and has a solid theoretical foundation. It can decompose a raw signal into a series of subseries with specific bandwidth in spectral domain, and demonstrate better noise robustness and more precise component separation [42]. Recently, this method has been successfully employed to solve various problems, such as biomedical signal processing [43], fault detection [44], financial time series analysis [45] and so on.

Considering the excellent characteristics of VMD in dealing with non-linear and irregular signal decomposition, and the advantageous performance of LSTM in prediction, in this paper, a new hybrid model based on VMD and LSTM is proposed for non-ferrous metals price forecasting. Through the hierarchical model, the instability of the original sequence and limited ability of single predictor processing complex series are handled. In detail, we first decompose the original price series into several relatively stable IMFs (Intrinsic Mode Functions) using VMD, and the intrinsic linear and non-linear characteristics of the data can be captured more accurately. Then we train the LSTM forecasting models for each IMF sequence and obtain the corresponding prediction results. Finally, the forecasting results of all IMF components are aggregated to produce an ensemble forecasting consequence for the original price series. Compared with traditional methods, the proposed method can capture the internal characteristics of the original data, learn the interdependence of historical data, and then effectively improve the overall prediction ability.

The main contributions of this paper are summarized as follows.

(1) A novel non-ferrous metal price forecasting framework is proposed by combining with VMD and LSTM network. VMD, as a competitive signal decomposition method, is utilized to capture the irregular characteristics of raw data, and has better adaptability and decomposition effect than EMD. The LSTM network, which is a deep learning method, can explore the hidden high-level interdependence of the non-ferrous metal price data, and the prediction is very effective.

Please cite this article as: Y. Liu, C. Yang, K. Huang et al., Non-ferrous metals price forecasting based on variational mode decomposition and LSTM network, Knowledge-Based Systems (2019) 105006, https://doi.org/10.1016/j.knosys.2019.105006.

(2) The ratio of residual energy after decomposition is subtly utilized to solve the problem of determining the appropriate mode number K in VMD. The appropriate K can not only improve the overall forecasting accuracy, but also avoid unnecessary computational overhead.

(3) Numerical experiments indicate that the proposed VMD-LSTM has higher level prediction accuracy and directional forecasting accuracy compared with other forecasting engines, and the comprehensive effect is better.

The remainder of this paper is organized as follows: Section 2 describes the basic methodology of VMD and LSTM, and introduces the system framework and forecasting procedure of the proposed hybrid model. In Section 3, the experimental design, the simulation results and comparative analysis are presented. Finally, Section 4 provides the conclusion remarks.

2. Methodology

In this section, two modules of the proposed hybrid method: variation mode decomposition (VMD) and long short-term memory (LSTM) network, as well as the general framework of VMD-LSTM model will be described detailedly.

2.1. Variational mode decomposition

Variational mode decomposition (VMD) developed by Dragomiretskiy and Zosso [41] in 2014 is a state-of-the-art multiresolution technique for adaptive and non-recursive signal processing. The decomposition process of VMD is the construction and solution of the variational problem. It can decompose a real valued signal into a series of band-limited Intrinsic Mode Function (IMF) u_k in spectral domain owning specific sparsity properties. It is assumed that each IMF u_k can be concentrated around a center frequency ω_k which is determined along with the decomposition process. To access the bandwidth of each u_k, the following three steps should be fulfilled:

Step 1: For each IMF u_k, adopting Hilbert transform to obtain a unilateral frequency spectrum.

Step 2: Shifting the IMF's frequency spectrum to baseband region by mixing with an exponential tuned to the respective estimated center frequency.

Step 3: Estimating each IMF's bandwidth by utilizing the Gaussian smoothness of the demodulated signal, i.e., the squared L2-norm of the gradient.

The objective function explores the K IMFs of original signal, of which the sum value of the estimated bandwidth is the minimum. Furthermore, the aggregation of the IMFs should be equal to the input signal $f(t)$, which is the constraint condition. Subsequently, the constrained variational problem can be given as follows:

$$\begin{aligned} &\min_{\{u_k\},\{\omega_k\}} \left\{ \sum_{k=1}^{K} \left\| \partial_t \left[(\delta(t) + \frac{j}{\pi t}) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \right\} \\ &s.t. \sum_{k=1}^{K} u_k = f(t) \end{aligned} \tag{1}$$

where $f(t)$ is the original signal to be decomposed, $\delta(t)$ is the Dirac distribution, t implicates the time script, $*$ denotes the convolution operator, u_k and ω_k represent the kth IMF and the corresponding center frequency respectively. $\{u_k\}$ stands for the set of IMFs $\{u_1, u_2, \ldots, u_K\}$ and $\{\omega_k\}$ denotes the set of center frequencies of the IMFs $\{\omega_1, \omega_2, \ldots, \omega_K\}$. K denotes the total number of the IMFs, additionally the IMF with high-order k denotes low-frequency component.

In order to translate the above constraint variational problem into an unconstrained optimization problem, both quadratic penalty term and Lagrangian multipliers are considered. The augmented Lagrange equation can be defined as:

$$\begin{aligned} L(\{u_k\}, \{\omega_k\}, \lambda) = \alpha \sum_{k=1}^{K} \left\| \partial_t \left[(\delta(t) + \frac{j}{\pi t}) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \\ + \left\| f(t) - \sum_{k=1}^{K} u_k(t) \right\|_2^2 + \left\langle \lambda(t), f(t) - \sum_{k=1}^{K} u_k(t) \right\rangle \end{aligned} \tag{2}$$

where $\lambda(t)$ is Lagrange multiplication factor for tightening restraint, α refers to the balance parameter of the data fidelity constraint, $\left\| f(t) - \sum_{k=1}^{K} u_k(t) \right\|_2^2$ is the quadratic penalty term for accelerating the convergence velocity.

To address this problem, the Alternate Direction Method of Multipliers (ADMM) [46] is adopted to find the saddle point of the augmented Lagrangian. In accordance with ADMM optimization approach, the u_k and ω_k can be updated in two directions to realize the analysis process of VMD, and the following iterative solutions are obtained:

$$u_k^{n+1} = \arg\min_{u_k} L\left(\left\{u_{i<k}^{n+1}\right\}, \left\{u_{i\geq k}^{n+1}\right\}, \left\{\omega_i^n\right\}, \left\{\lambda^n\right\}\right) \tag{3}$$

$$\omega_k^{n+1} = \arg\min_{\omega_k} L\left(\left\{u_i^{n+1}\right\}, \left\{\omega_{i<k}^{n+1}\right\}, \left\{\omega_{i\geq k}^{n}\right\}, \left\{\lambda^n\right\}\right) \tag{4}$$

$$\lambda^{n+1} = \lambda^n + \tau \left(f(t) - \sum_{k=1}^{K} u_k^{n+1} \right) \tag{5}$$

the condition of convergence is $\sum_{k=1}^{K} \left\| u_k^{n+1} - u_k^n \right\|_2^2 \Big/ \left\| u_k^n \right\|_2^2 < \varepsilon$, where n represents the number of iterations and τ is the time step of the dual ascent. Consequently, the solutions for u_k and ω_k are expressed as follows:

$$\hat{u}_k^{n+1} = \frac{\hat{f}(\omega) - \sum_{i\neq k} \hat{u}_i(\omega) + \frac{\hat{\lambda}(\omega)}{2}}{1 + 2\alpha(\omega - \omega_k)^2} \tag{6}$$

$$\omega_k^{n+1} = \frac{\int_0^\infty \omega \left| \hat{u}_k^{n+1}(\omega) \right|^2 d\omega}{\int_0^\infty \left| \hat{u}_k^{n+1}(\omega) \right|^2 d\omega} \tag{7}$$

where $\hat{f}(\omega)$, $\hat{\lambda}(\omega)$, $\hat{u}_i(\omega)$, $\hat{u}_k^{n+1}$ are the Fourier transforms of $f(\omega)$, $\lambda(\omega)$, $u_i(\omega)$, u_k^{n+1} respectively. The complete procedure of VMD can be illustrated in Fig. 1.

2.2. Long short-term memory network

LSTM network is an extended version of recurrent neural network (RNN) architecture, and has been widely applied in various fields. It contains not only the external loop, but also the internal circulation in the memory cells.

2.2.1. Recurrent neural network

RNN is a neural network model which specializes in dealing with the non-linear time series prediction problem [47]. The network structure of RNN is similar to that of multilayer perception, whereas it employs cyclical or recurrent connections which use previous hidden node state as input, as shown in Fig. 2. The internal connections between hidden nodes enable information to travel forward or backward, so the output of a single RNN cell can be described as a function of previous RNN state and the input layer. In principle, the recurrent architecture can map the whole historical input data to every output, and this plays a crucial role in the accurate prediction of time sequences.

Despite the good performance that traditional RNN exhibits in complex time series modeling problem, there is still a fly in

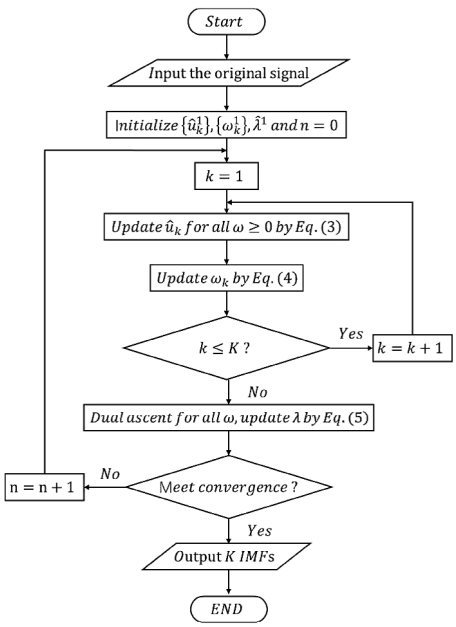

Fig. 1. The complete procedure of VMD.

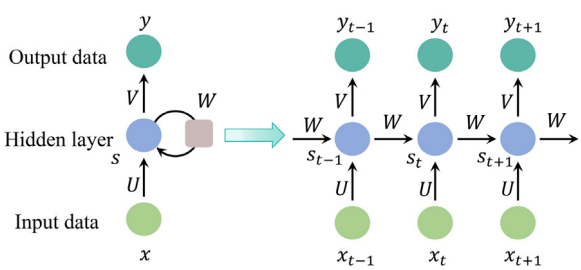

Fig. 2. The schematic diagram of the architecture of RNN.

the ointment. On one hand, it is difficult to find the best time lag namely time window size automatically to learn the temporal series processing. On the other hand, due to the long chains of operators introduced by recurrent connections, the gradient of RNN may decay exponentially or grow catastrophically during training when handing long-term dependencies of data, leading to the well-known vanishing gradients problem and exploring gradients problem. In consequence, traditional simple RNN may not be the ideal method for long-term forecasting.

2.2.2. Long short-term memory network

To overcome the aforementioned shortcomings of traditional RNN, LSTM network was proposed by Hochreiter and Schmidhuber in 1997 [19]. LSTM network is a deformation of RNN, but it introduces gating mechanism and replaces nodes in the RNN with special units called memory blocks. The memory blocks not only contain memory cell that can memorize the temporal state, but also have special adaptive multiplicative units called gates to control the information flows in the block.

The basic structure of a LSTM memory block is exhibited in Fig. 3. The memory cell essentially acts as an accumulator of the state information, and each memory block contains three types of valves: input gate, forget gate and output gate. The input gate i_t determines how much new information will be added to the current cell state. The forget gate f_t controls how much information will be forgotten from the previous cell state, and it prevents the internal cell values growing without boundaries.

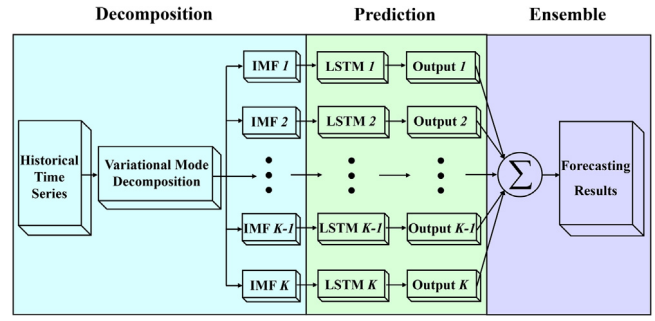

Fig. 3. The basic structure of LSTM memory block.

The output gate o_t is designed to decide what information will be output by filtering the new state.

A LSTM network computes a mapping from a historical input sequence x to a predicted output sequence $y = (y_1, y_2, \ldots, y_T)$ by using the following steps iteratively from $t = 1$ to T, where T is the prediction period.

First, input gate decides what new information should be stored in the cell state, as well as creates new candidate value $\widehat{C}_t$ that could be possibly added to the state.

$$i_t = \sigma\left(W_i \cdot [h_{t-1}, x_t] + b_i\right) \tag{8}$$

$$\widehat{C}_t = \tanh\left(W_c \cdot [h_{t-1}, x_t] + b_c\right) \tag{9}$$

Next, forget gate decides what information should be got rid of the cell state.

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, x_t] + b_f\right) \tag{10}$$

Via throwing away part of the information from the old cell and adding filter candidate value, the old cell state C_{t-1} is updated into new cell state C_t.

$$C_t = f_t \odot C_{t-1} + i_t \odot \widehat{C}_t \tag{11}$$

Lastly, output gate filters the cell state and calculates the desired information, and the final output y_t is obtained as follows

$$o_t = \sigma\left(W_o \cdot [h_{t-1}, x_t] + b_o\right) \tag{12}$$

$$h_t = o_t \odot \tanh\left(C_t\right) \tag{13}$$

$$y_t = \phi(W_y h_t + b_y) \tag{14}$$

where C_t is the cell state and $\widehat{C}_t$ denotes the updated new state, h_t is the hidden layer state namely the activation of the memory block. W_i, W_f, W_c, W_o and W_y denote the appropriate weight matrices, b_i, b_f, b_c, b_o and b_y represent the corresponding bias vectors, and they can be determined by the Back-Propagation algorithm. In addition, $\sigma(\ )$ and tanh( ) are Sigmoid and Hyperbolic tangent function respectively, $\odot$ is the element-wise product of the vectors and ϕ is the network output activation function.

2.2.3. VMD-LSTM ensemble paradigm

In view of the merits of VMD and LSTM, in this paper, VMD method and LSTM model are combined to establish a novel hybrid forecasting model based on the concept of "Decomposition and Ensemble". The proposed hybrid model is short as VMD-LSTM. Suppose $f(t), t = 1, 2, \ldots, M$ is the original time series, M is the sample number, VMD decomposes the original time series into several interdependent components $u_i(t), i = 1, 2, \ldots, K$, K is the total number of components. The decomposition can be helpful for precise modeling of the intermittent characteristics and improving the forecasting accuracy. As the prediction core of the proposed forecasting engine, LSTM completes the

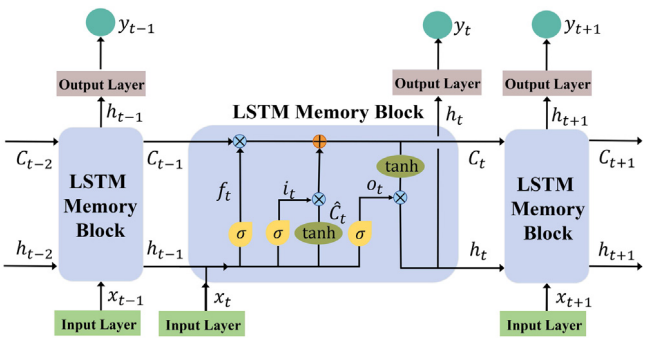

Fig. 4. The schematic description of the hybrid VMD-LSTM model.

modeling and forecasting for each component obtained by decomposition. The schematic description of the hybrid VMD-LSTM model is presented in Fig. 4. In details, the complete procedure of the proposed VMD-LSTM framework is conducted and given in Algorithm 1.

Algorithm 1 VMD-LSTM in price forecasting.

Input: The historical price data of non-ferrous metal $f(t)$;
Output: The forecasted price of non-ferrous metal $y(t)$;
#Decomposition
Step 1: Randomly give the mode number K, calculate λ, $\hat{u}_k$, ω_k according to the algorithm flow of VMD shown in Fig. 1 and Eqs. (5)–(7), and obtain the corresponding IMFs of original signal $f(t)$.
Step 2: Calculate the residual energy of the decomposition on the basis of Eq. (19), and determine the appropriate mode number K to obtain the final decomposition results $u_i(t), i = 1, \ldots, K$.
#Modeling
Step 3: Split each component $u_i(t)$ according to step size to get input and output for training model.
Step 4: For each component, LSTM network is trained with the data obtained in Step 3 to general a forecasting model.
#Forecasting
Step 5: For each component, input test data into the forecasting model, and get individual forecasted result $y_i(t), i = 1, \ldots, K$.
#Ensemble
Step 6: Aggregated the forecasting results of each component to generate the final forecasted result $y(t) = \sum_{i=1}^{K} y_i(t)$.

3. Experimental results and comparative analysis

3.1. Data description

This paper adopts three datasets of London Metal Exchange (LME) to evaluate the performance of the proposed model, and the data are future closing prices obtained from the website of LME (https://www.lme.com/Metals/Non-ferrous). The price time series of Zinc, Copper and Aluminum are illustrated in Fig. 5. Due to the different features of different metal markets, these three data series have remarkably different fluctuation characteristics, therefore these three data series adopted in this paper can comprehensively and systematically evaluate the effectiveness and practicability of the proposed model. In particular, the time span for each price is from June 02, 2006 to March 21, 2019 excluding public holidays, with a total of 3248 daily observations. The data from June 02, 2006 to December 30, 2016 is used for model training (2862 observations), and the remainder is used as the testing set (566 observations) to verify the effectiveness of the proposed forecasting model based on the comprehensive error evaluation criteria.

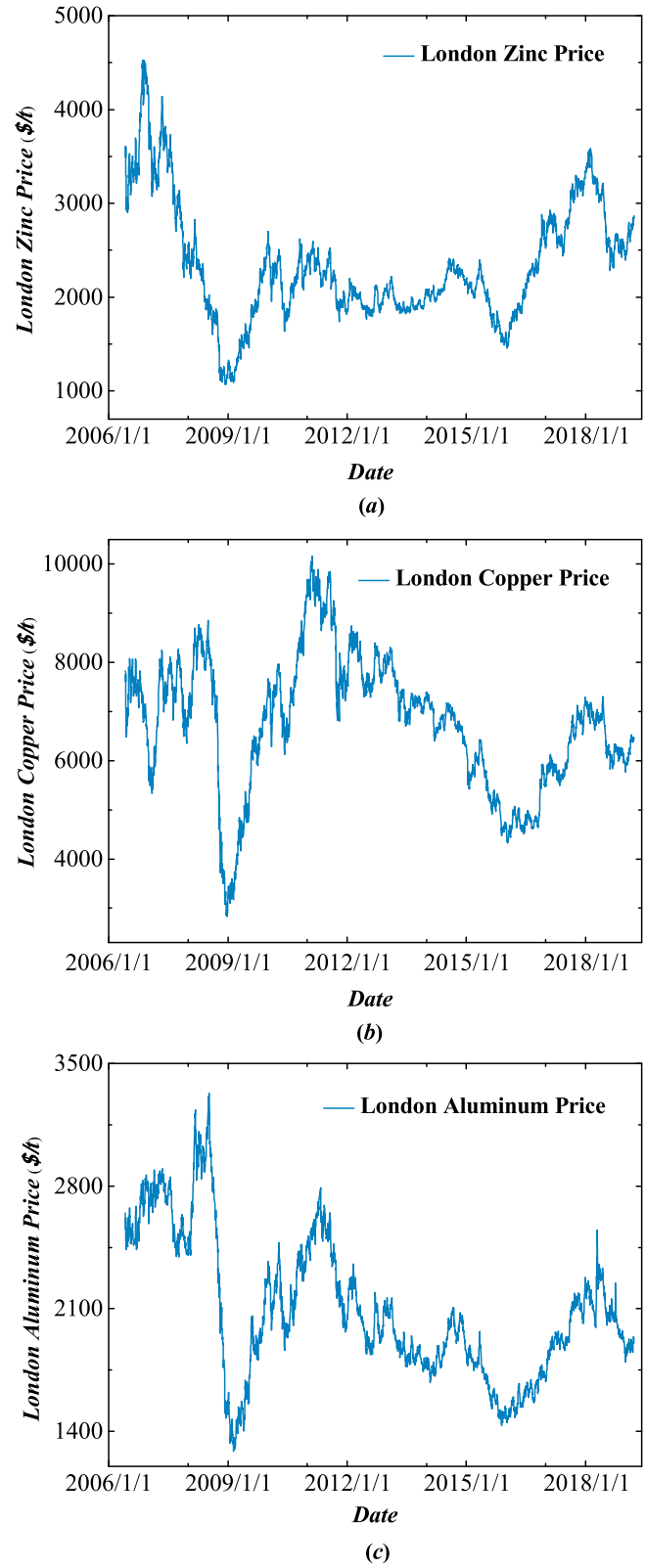

Fig. 5. The non-ferrous metals price on the London Metal Exchange from 2nd June 2006 to 21th March 2019. (a) Zinc futures price. (b) Copper futures price. (c) Aluminum futures price.

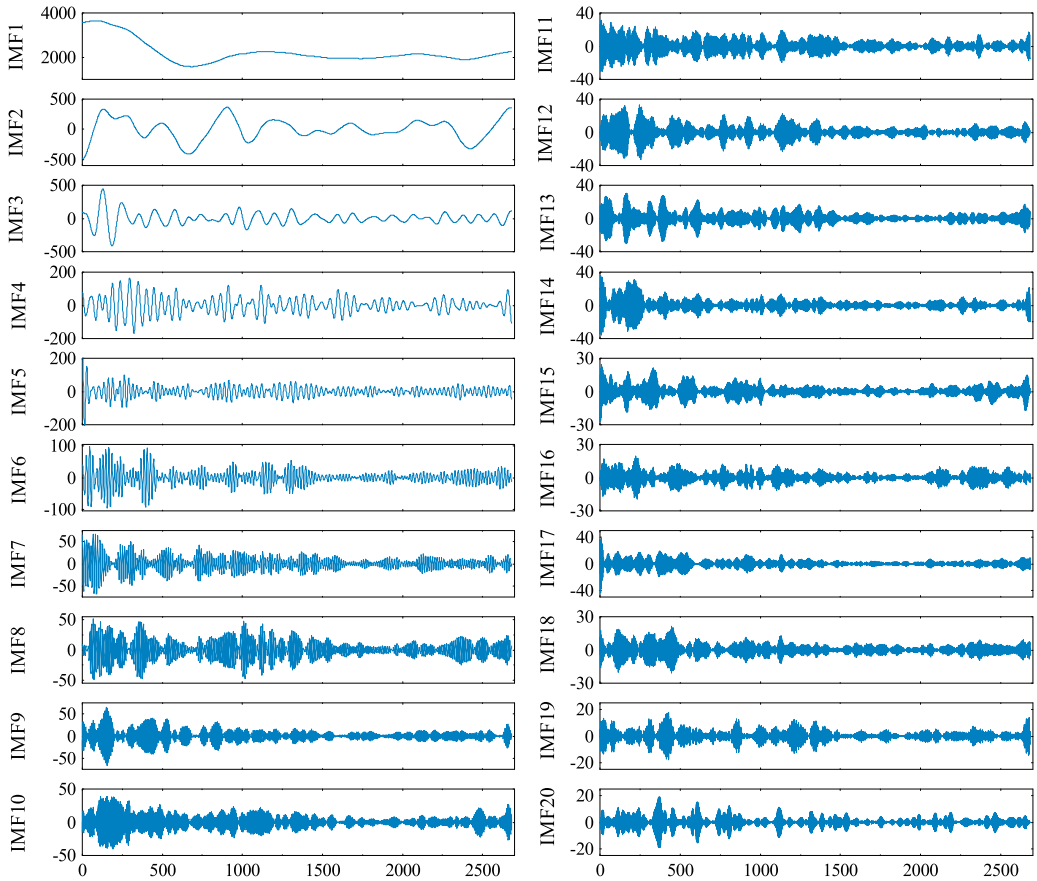

Fig. 6. The data decomposition results of LME Zinc futures price.

3.2. Performance evaluation criteria

It is important to apply multiple performance measures when assessing the forecasting ability of the developed model. This paper adopts two types of criteria for evaluation of level prediction and directional forecasting. For level prediction accuracy, MAE, MAPE and RMSE are utilized. Among the three statistical measures, MAE is the mean absolute error, MAPE is the mean absolute percentage error, and RMSE is the root mean squared error. The detailed computational formulas are respectively expressed as follows:

$$MAE = \frac{1}{m}\sum_{t=1}^{m}|R_t - P_t| \tag{15}$$

$$MAPE = \frac{1}{m}\sum_{t=1}^{m}\left|\frac{R_t - P_t}{R_t}\right| \tag{16}$$

$$RMSE = \sqrt{\frac{1}{m}\sum_{t=1}^{m}(R_t - P_t)^2} \tag{17}$$

where R_t represents the real value for timestamp t, P_t represents the corresponding predicted value, and m is the number of predicted results. Apparently, smaller values of these criteria indicate better forecasting performance.

For directional forecasting accuracy, the capacity of predicting correct direction is more important from the economical point of view. And it can be measured by directional statistic D_{stat}, typically defined by Eq. (18) :

$$D_{stat} = \frac{1}{m}\sum_{t=1}^{m} a_t \times 100\%$$
$$a_t = \begin{cases} 1, (R_{t+1} - R_t)(P_{t+1} - R_t) \geq 0 \\ 0, otherwise \end{cases} \tag{18}$$

3.3. Data decomposition and parameter setting

In the proposed VMD-LSTM model, VMD is executed to disaggregate the raw price series into several modes to reduce the non-stationary and non-linear characteristics, thus makes them easy to be predicted. Before the implementation of decomposition by VMD, the number of components K needs to be predetermined. If K is too small, the original series may not be decomposed sufficiently and it causes inaccurate prediction. On the other side, if the signal is over-decomposed, the difference between each component becomes very small, and it gives rise to accuracy reduction and unnecessary computing overheads. To identify the optimal mode number, the ratio of residual energy r_{res} to the original signal energy is introduced. Here, the r_{res} can be defined as follows:

$$r_{res} = \frac{1}{M}\sum_{t=1}^{M}\left|\frac{f(t) - \sum_{k=1}^{K} u_k(t)}{f(t)}\right| \tag{19}$$

where $f(t)$ is the original price series, $u_k(t)$ is the individual component, and M is the sample number. Since r_{res} is residual error after decomposition, hence it can be applied as an optimization index in the VMD procedure. Empirically, when r_{res} is less than 1% and there is no significant downward trend, the component

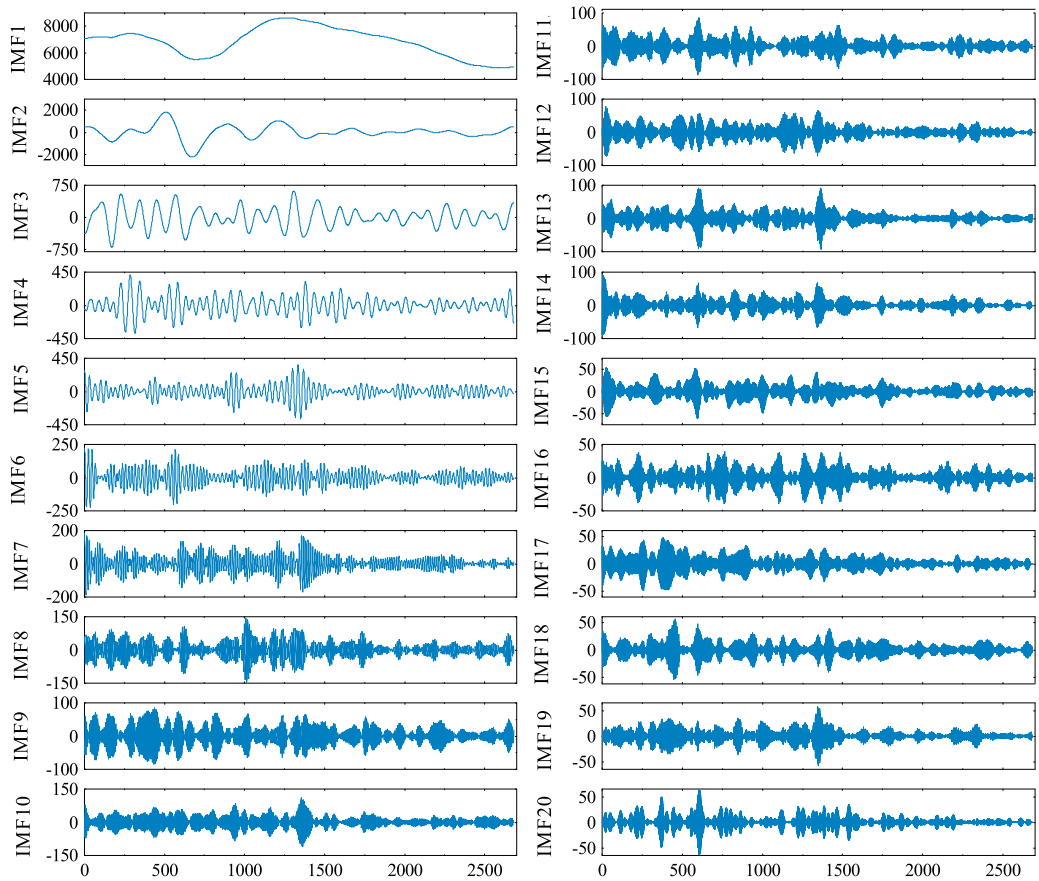

Fig. 7. The data decomposition results of LME Copper futures price.

Table 1
The ratio of residual energy corresponding to different K.

K	Zinc price series (r_{res},%)	Copper price series (r_{res},%)	Aluminum price series (r_{res},%)
5	0.95	0.79	0.69
6	0.77	0.74	0.62
7	0.67	0.63	0.51
8	0.62	0.54	0.45
9	0.53	0.47	0.39
10	0.48	0.42	0.34
11	0.42	0.36	0.30
12	0.36	0.31	0.25
13	0.32	0.28	0.20
14	0.27	0.24	0.18
15	0.24	0.21	0.18
16	0.21	0.19	0.14
17	0.20	0.18	0.13
18	0.18	0.15	0.12
19	0.15	0.14	0.11
20	**0.13**	**0.12**	**0.10**
21	0.13	0.12	0.09
22	0.12	0.11	0.09

number can be determined. The residual energy under different K for different price series is shown in Table 1.

As shown in Table 1, all r_{res} are below 1%. Moreover, when K is greater than 20, the trend of descending inclines to be stable. Taking into account the computational cost, the suitable value of K is taken as 20. Applying VMD to the price series, the subseries of different data are shown in Figs. 6 to 8 respectively. Apparently, these extracted subsequences are smoother and more regular than the original data, which is conductive to the improvement of the forecasting accuracy. The subseries with relatively lower frequency represent the whole tendency of the raw price series, and the higher frequency subseries reflect the local volatility tendency.

After the decomposition of price series, the next task is to construct a forecasting model for each subsequence generated by VMD. In this work, LSTM network is employed for the prediction of each subsequence. We use Keras to build LSTM network, which consists of an input layer, a LSTM layer and an output layer. For the input layer, the step size is set as 30, namely using the data from the first 30 days to predict the price of next day. The number of hidden layer neurons is determined as 256, by trial-and-error method, and the sigmoid function is chosen as the activation function. The learning rate and the batch size are set as the default value of 0.01 and 8, and the maximum number of training equals to 100.

For the purpose of analyzing the prediction performance of the VMD-LSTM model, we compare it with other baseline models, such as ARIMA, FNN, LSTM, EMD-FNN, EMD-LSTM and VMD-FNN. For ARIMA model, the optimal values of parameter p, d, q are determined based on the Akaike's Information Criteria (AIC). As to FNN (I-H-O), there is one neuron in each of the input and output layer. By the grid search (GS) algorithm, the hidden nodes H is set to 100, and the model runs 5000 times iteratively with the same learning rate of 0.01 for training. It should be noted that all of the simulations are implemented in Python 3.x running on an Intel@ Core™ i7-7700HQ processor operating at 3.40 GHz with a 16.00 GB installed RAM.

3.4. Empirical results and analysis

Some experiment results are presented in this section to verify the efficiency of the proposed VMD-LSTM model. Based on the

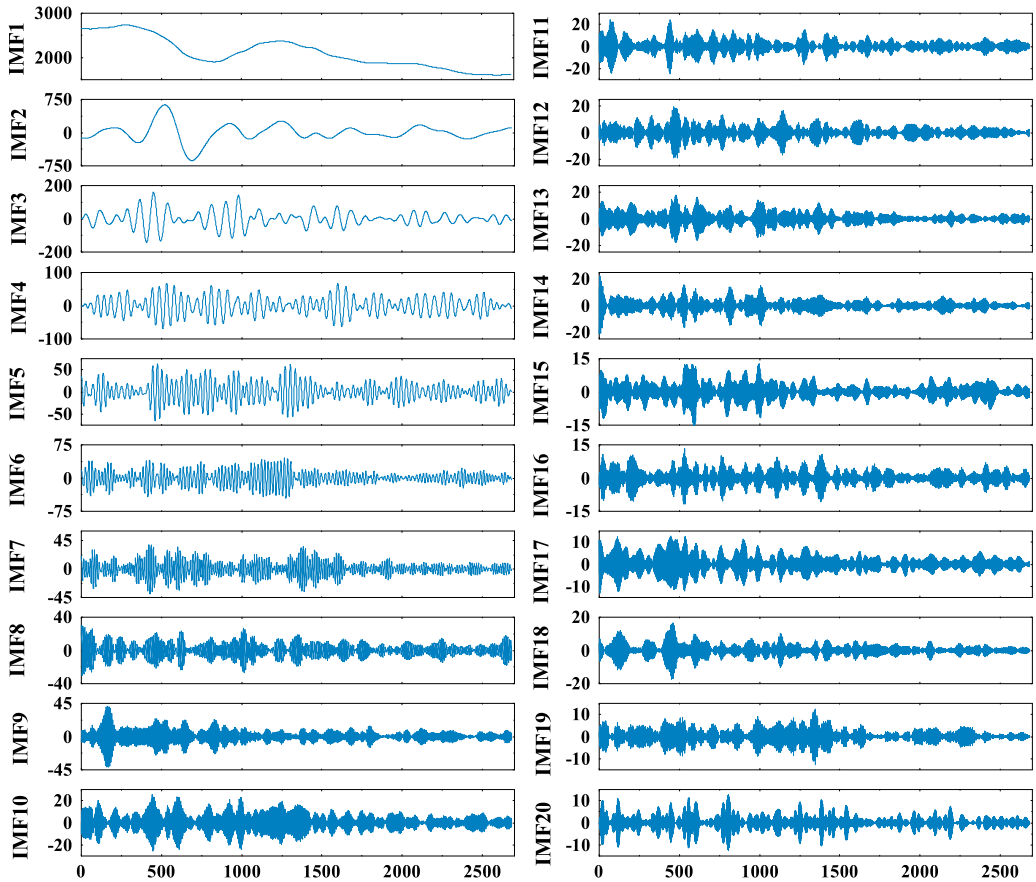

Fig. 8. The data decomposition results of LME Aluminum futures price.

above settings, testing experiments are conducted on three typical non-ferrous metals price data of Zinc, Copper and Aluminum. In the following, the accuracy evaluation indices are used to analyze the forecasting performance of these models. Figs. 9 to 12 display the evaluation criteria comparison of different forecasting models for Zinc, Copper and Aluminum. Take Zinc price as example, it is clearly seen from the histograms that, regardless of level prediction indices (MAE, MAPE, RMSE) or the directional forecasting index (D_{stat}), the VMD-LSTM model has the best prediction performance among all forecasting models. In particular, for the typical single model forecasting methods, they perform similarly in deviation error of the prediction results, but ARIMA model has a higher accuracy in direction prediction. Furthermore, compared with single model methods, hybrid models based on "Decomposition-and-Ensemble" framework have excellent prediction result, and they can improve the forecasting performance of single model considerably.

Besides that, in the hybrid models, the VMD-based prediction models outperform the models which use EMD decomposition method. It can be concluded that the VMD decomposition technique is superior to EMD. After decomposition, the hybrid models which utilize LSTM network as the prediction core generally outperform those with FNN as the predictor, and the superiorities of LSTM network for time series processing are reflected. As expected, the evaluations of Copper and Aluminum prices show a similar situation, and the methods considered have certain robustness and applicability.

For a quantitative analysis of the prediction results, the evaluation indices of different models aforementioned for different price data are listed in Tables 2 to 4. Similarly, the price of Zinc is discussed as an example. The deviation error evaluation criteria MAE, MAPE, RMSE of VMD-LSTM model are 3.77, 0.11% and 4.79

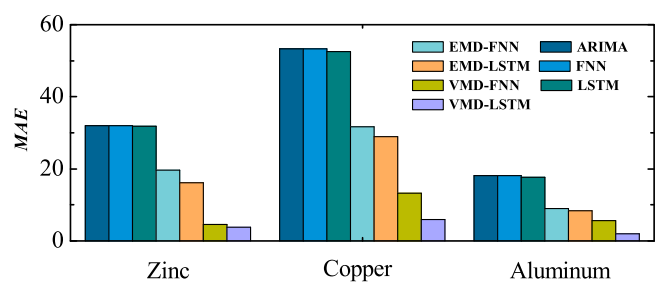

Fig. 9. MAE comparisons of different models for different price data.

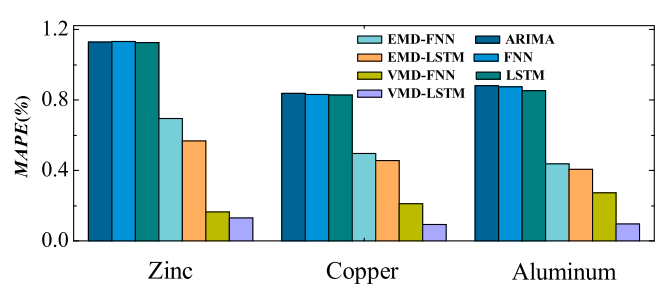

Fig. 10. MAPE comparison of different models for different price data.

respectively, which are significantly smaller than those of other forecasting models. As for the accuracy of direction prediction, the value of D_{stat} of VMD-LSTM is 96.26% which is the largest among all models, namely the VMD-LSTM model has the best forecasting effect. For the three single model prediction methods

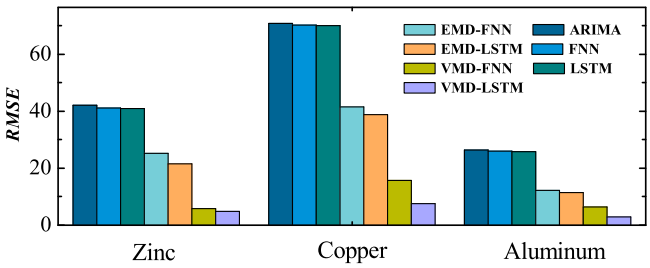

Fig. 11. RMSE comparisons of different models for different price data.

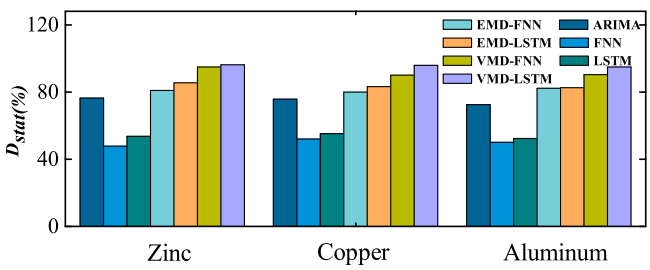

Fig. 12. D_{stat} comparisons of different models for different price data.

of ARIMA, FNN and LSTM, since the original price series fluctuate drastically, vary widely and change irregularly as illustrated in Fig. 5, the complex characteristics such as non-linearity and non-stationarity are very obvious, and the single model methods generally do not perform well. They perform similarly on the three deviation error criteria of MAE, MAPE and RMSE. However, on D_{stat}, the values of ARIMA, FNN and LSTM are 76.28%, 47.87% and 53.72% respectively. The performance of ARIMA is better than those based on neural network.

Furthermore, by comparing FNN with EMD-FNN and VMD-FNN, LSTM with EMD-LSTM and VMD-LSTM, it is shown that hybrid models are superior to single model prediction methods in all cases. This may be ascribed to the fact that since the original signal fluctuates violently and varies dramatically, it is difficult to extract and analyze the variation law by using a single model directly, and it is usually hard to obtain satisfactory forecasting results. After the mode decomposition of the original signal, the decomposed subsequences are more stable and the period is relatively more regular. The interference and coupling between different scales feature information in the original data are reduced, which makes it easier to obtain the complex internal characteristics of the original series, including both linear and non-linear characteristics, thus lightening the burden and difficulty of the prediction models. Hence, the forecasting performance of the hybrid model based on decomposition is considerably improved compared to the single model approach.

In the hybrid models, no matter what the prediction core is in the model, it is clearly that the VMD-based models outperform the EMD-based models. The possible reasons for the better performance of VMD could be summarized in the following three aspects: (1) VMD is less sensitive to noisy data unlike EMD decomposition method; (2) VMD has better decomposition adaptability, and the residual noise in each component is very small, which can provide deeper decomposition subseries; (3) Compared with EMD, VMD is capable of obtaining a large number of similar frequency subseries, and it can reveal the characteristics of each component more integrally. Overall, VMD overcomes the problems of model aliasing, noise sensitivity, boundary effects, etc. in EMD, and has a better decomposition effect, so the improvement of prediction accuracy is more remarkable.

Table 2
Error evaluations of different models for LME Zinc price.

Models	MAE	MAPE(%)	RMSE	D_{stat}(%)
ARIMA	31.96	1.13	41.07	76.28
FNN	32.02	1.13	41.06	47.87
LSTM	31.80	1.12	40.89	53.72
EMD-FNN	19.69	0.69	25.13	80.93
EMD-LSTM	16.13	0.57	21.54	85.54
VMD-FNN	4.59	0.16	5.91	95.90
VMD-LSTM	**3.77**	**0.11**	**4.79**	**96.26**

Table 3
Error evaluations of different models for LME Copper price.

Models	MAE	MAPE(%)	RMSE	D_{stat}(%)
ARIMA	53.23	0.84	70.76	75.75
FNN	52.71	0.83	70.31	52.13
LSTM	52.60	0.83	70.10	55.32
EMD-FNN	31.61	0.49	41.46	79.86
EMD-LSTM	28.91	0.46	38.82	83.21
VMD-FNN	13.18	0.21	15.78	90.02
VMD-LSTM	**5.90**	**0.09**	**7.48**	**95.89**

Table 4
Error evaluations of different models for LME Aluminum price.

Models	MAE	MAPE(%)	RMSE	D_{stat}(%)
ARIMA	18.17	0.88	26.33	72.57
FNN	18.07	0.88	25.99	50.18
LSTM	17.69	0.86	25.79	52.30
EMD-FNN	9.05	0.44	12.16	82.17
EMD-LSTM	8.42	0.41	11.47	82.57
VMD-FNN	5.60	0.27	6.45	90.18
VMD-LSTM	**2.01**	**0.10**	**2.83**	**94.77**

For the predictor, as can be seen from Table 2, LSTM has certain advantages over FNN no matter which decomposition method is used, and it may be attributed to the following reasons: For traditional FNN, the key restriction affecting its prediction performance is the acyclic nature. All inputs and outputs are independent of each other, and this is not ideal for modeling time series that are highly correlated. In addition, FNN is greatly influenced by hyperparameters, and is prone to local minimum and overfitting, so the generalization ability is usual poor [48]. LSTM is one of the most powerful models for processing temporal data, and it replaces the hidden layer neurons in FNN with memory cells and forms the structure of information circulation. Therefore, LSTM can effectively associate the historical information with current input data, thereby can capture the dynamic characteristics of data with high accuracy. Because VMD and LSTM have unique advantages in signal decomposition and time series processing, based on the successful combination of the two algorithms, the proposed VMD-LSTM model has the best forecasting results among all the models considered. As expected, the aforementioned change laws are basically the same in the price predictions of Copper and Aluminum, reflecting the robustness of the prediction models.

In order to show the forecasting effects of different models more intuitively, Fig. 13 describes the prediction of the prices of Zinc, Copper and Aluminum. Since the price series of non-ferrous metals fluctuate widely as presented in Fig. 5, in order to increase the discrimination of different models to have a better representation, we analyze the prediction error between the predicted value and the real value of each model. Fig. 13 shows the prediction error of all hybrid models and typical single model LSTM from January 01, 2017 to March 31, 2017. Specially, the prediction error of single model varies dramatically with a wide range of fluctuations, and the forecasting accuracy is significantly lower than those of the hybrid prediction models. After decomposition,

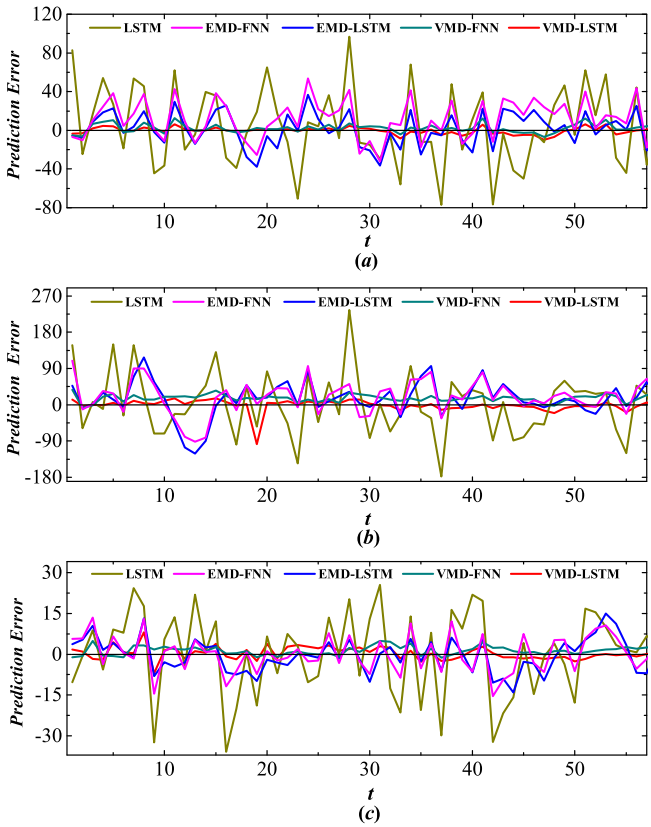

Fig. 13. The prediction error of different forecasting models for different price data. The prediction error is the difference between the predicted value and the real value. The smaller the prediction error is, the better the prediction effect of the model is. (a) The prediction error for LME Zinc price. (b) The prediction error for LME Copper price. (c) The prediction error for LME Aluminum price.

the prediction models have a significant forecasting promotion. Although the price fluctuations and variations of different metal prices vary, as can be seen from the figures, all the curves of predicted results of hybrid models are roughly approximating the real values. In addition, the prediction errors of VMD-based models are smaller than those of the EMD-based method, and LSTM network has certain advantages over FNN. For all price series, the prediction errors of VMD-LSTM are basically distributed around zero with a small scale, namely the proposed model has a satisfactory forecasting result.

In particular, the runtime of all forecasting models on different data is shown in Table 5 to facilitate analysis of the computational complexity [49,50] of the algorithms. All results are average over 10 interdependent realizations. As can be seen from the table, the runtime of LSTM in the single model methods is relatively long for all price data, and the computational complexity of decomposition-based hybrid forecasting model is higher than that of the single model. Obviously, the proposed method has the largest computational cost among the models considered as the average runtime of a VMD-LSTM model is about 21 s. Nevertheless, the real-time demand of price forecasting in practical applications is not high, the computational cost will rarely be prioritized. On the other hand, with the development of hardware computing power as well as the application of parallel computing technology, the runtime of the proposed method will be decrease in the future.

Table 5
The runtime of different forecasting models for different price data.

Models	Zinc(s)	Copper(s)	Aluminum(s)
ARIMA	0.003	0.003	0.003
FNN	0.037	0.038	0.036
LSTM	0.228	0.230	0.225
EMD-FNN	0.948	0.991	1.041
EMD-LSTM	4.672	4.678	4.788
VMD-FNN	14.036	15.097	14.220
VMD-LSTM	20.559	20.617	20.689

4. Conclusion

Non-ferrous metals are pivotal industrial raw materials, and the changes in price impact the operation of enterprises even the national economy. In an attempt to improve the forecasting accuracy of non-ferrous metals price, this paper proposed a novel VMD-LSTM forecasting model. Here, the VMD method as a powerful signal processing technique is utilized to decompose the original price series into a group of subseries for reducing the instability of the raw data, the LSTM network is employed to build forecasting models for each independent component, and the individual forecasting results of each subseries are aggregated to an ensemble result. To investigate the forecasting performance of the proposed model, testing experiments are carried out on the daily data of Zinc, Copper and Aluminum from LME, and ARIMA, FNN, LSTM, EMD-FNN, EMD-LSTM and VMD-FNN are utilized for comparison. The qualitative and quantitative results demonstrate that the proposed VMD-LSTM model outperforms other state-of-the-art methods in all evaluation criteria. Since the proposed method takes the irregular feature and interdependence of time-series data into account, it is a promising methodology and can be extensively applied to other forecasting fields.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China (Grants No. 61703439, 61860206014), in part by the Foundation for Innovative Research Groups of the National Natural Science Foundation of China (Grant No. 61621062), in part by the Innovation-Driven Project of Central South University, China (2019CX020), in part by Fundamental Research Funds for the Central Universities of Central South University, China (2019zzts574) and in part by the 111 Project, China (B17048).

References

[1] Meirui Zhong, Ruifang He, Jinyu Chen, Jianbai Huang, Time-varying effects of international nonferrous metal price shocks on China's industrial economy, Physica A 528 (2019) 121299.
[2] Juan Wang, Mingming Hu, João FD Rodrigues, The evolution and driving forces of industrial aggregate energy intensity in China: An extended decomposition analysis, Appl. Energy 228 (2018) 2195–2206.
[3] Clinton Watkins, Michael McAleer, Econometric modelling of non-ferrous metal prices, J. Econ. Surv. 18 (5) (2004) 651–701.
[4] Antonio Gargano, Allan Timmermann, Forecasting commodity price indexes using macroeconomic and financial predictors, Int. J. Forecast. 30 (3) (2014) 825–843.
[5] Huijia Li, Zhan Bu, Zhen Wang, Jie Cao, Yong Shi, Enhance the performance of network computation by a tunable weighting strategy, IEEE Trans. Emerg. Topics Comput. Intell. 2 (3) (2018) 214–223.
[6] Huijia Li, Zhan Bu, Aihua Li, Zhidong Liu, Yong Shi, Fast and accurate mining the community structure: Integrating center locating and membership optimization, IEEE Trans. Knowl. Data Eng. 28 (9) (2016) 2349–2362.
[7] Lihui Jia, Lin Wang, Multi-scale asynchronous belief percolation model on multiplex networks, New J. Phys. 21 (1) (2016) 015005.
[8] Rahul Sharma, Anurag Saxena, Kamal Vagrecha, Supply chain optimization of zinc industry: Opportunities, strategies and challenges., Global J. Enterprise Inf. Syst. 7 (3) (2015) 62–70.

[9] Gillian Dooley, Helena Lenihan, An assessment of time series methods in metal price forecasting, Resour. Policy 30 (3) (2005) 208–217.
[10] Goodness Aye, Rangan Gupta, Shawkat Hammoudeh, Won Joong Kim, Forecasting the price of gold using dynamic model averaging, Int. Rev. Financ. Anal. 41 (2015) 257–266.
[11] Yanhui Chen, Kaijian He, Chuan Zhang, A novel grey wave forecasting method for predicting metal prices, Resour. Policy 49 (2016) 323–331.
[12] Hesam Dehghani, Dejan Bogdanovic, Copper price estimation using bat algorithm, Resour. Policy 55 (2018) 55–61.
[13] Chang Liu, Zhenhua Hu, Yan Li, Shaojun Liu, Forecasting copper prices by decision tree learning, Resour. Policy 52 (2017) 427–434.
[14] Raúl Carrasco, Gabriel Astudillo, Ismael Soto, Máx Chacon, Diego Fuentealba, Forecast of copper price series using vector support machines, in: 2018 7th International Conference on Industrial Technology and Management, ICITM, IEEE, 2018, pp. 380–384.
[15] AG Malliaris, Mary Malliaris, Time series and neural networks comparison on gold, oil and the euro, in: 2009 International Joint Conference on Neural Networks, IEEE, 2009, pp. 1961–1967.
[16] Kumar Chandar Sivalingam, Sumathi Mahendran, Sivanandam Natarajan, Forecasting gold prices based on extreme learning machine, Int. J. Comput. Commun. Control 11 (3) (2016) 372–380.
[17] Fengyi Zhang, Zhigao Liao, Gold price forecasting based on RBF neural network and hybrid fuzzy clustering algorithm, in: Proceedings of the Seventh International Conference on Management Science and Engineering Management, Springer, 2014, pp. 73–84.
[18] Fernando Sánchez Lasheras, Francisco Javier de Cos Juez, Ana Suárez Sánchez, Alicja Krzemień, Pedro Riesgo Fernández, Forecasting the COMEX copper spot price by means of neural networks and ARIMA models, Resour. Policy 45 (2015) 37–43.
[19] Sepp Hochreiter, Jürgen Schmidhuber, Long short-term memory, Neural Comput. 9 (8) (1997) 1735–1780.
[20] Alex Graves, Abdel-rahman Mohamed, Geoffrey Hinton, Speech recognition with deep recurrent neural networks, in: 2013 IEEE International Conference on Acoustics, Speech and Signal Processing, IEEE, 2013, pp. 6645–6649.
[21] Hongbing Wang, Zhengping Yang, Qi Yu, Tianjing Hong, Xin Lin, Online reliability time series prediction via convolutional neural network and long short term memory for service-oriented systems, Knowl.-Based Syst. 159 (2018) 132–147.
[22] Martin Wöllmer, Moritz Kaiser, Florian Eyben, Björn Schuller, Gerhard Rigoll, LSTM-modeling of continuous emotions in an audiovisual affect recognition framework, Image Vis. Comput. 31 (2) (2013) 153–163.
[23] Xue-hong Zhu, Hong-wei Zhang, Mei-rui Zhong, Volatility forecasting in chinese nonferrous metals futures market, Trans. Nonferr. Met. Soc. China 27 (5) (2017) 1206–1214.
[24] Esteban Hernández, et al., Volatility of main metals forecasted by a hybrid ANN-GARCH model with regressors, Expert Syst. Appl. 84 (2017) 290–300.
[25] Mehdi Khashei, Mehdi Bijari, Gholam Ali Raissi Ardali, Improvement of auto-regressive integrated moving average models using fuzzy logic and artificial neural networks (ANNs), Neurocomputing 72 (4–6) (2009) 956–967.
[26] Bai Li, Research on WNN modeling for gold price forecasting based on improved artificial bee colony algorithm, Intell. Neurosci. 2014 (2014) 2:2.
[27] Zakaria Alameer, Mohamed Abd Elaziz, Ahmed A Ewees, Haiwang Ye, Zhang Jianhua, Forecasting copper prices using hybrid adaptive neuro-Fuzzy inference system and genetic algorithms, Nat. Resour. Res. (2019) 1–17.
[28] Diego García, Werner Kristjanpoller, An adaptive forecasting approach for copper price volatility through hybrid and non-hybrid models, Appl. Soft Comput. 74 (2019) 466–478.
[29] Hongwei Zhang, Xuehong Zhu, Yaoqi Guo, Haibo Liu, A separate reduced-form volatility forecasting model for nonferrous metal market: Evidence from copper and aluminum, J. Forecast. 37 (7) (2018) 754–766.
[30] Lean Yu, Shouyang Wang, Kin Keung Lai, Forecasting crude oil price with an EMD-based neural network ensemble learning paradigm, Energy Econ. 30 (5) (2008) 2623–2635.
[31] Gang Song, Qun Dai, A novel double deep ELMs ensemble system for time series forecasting, Knowl.-Based Syst. 134 (2017) 31–49.
[32] Wang Jun, Tang Lingyu, Luo Yuyan, Ge Peng, A weighted EMD-based prediction model based on TOPSIS and feed forward neural network for noised time series, Knowl.-Based Syst. 132 (2017) 167–178.
[33] Jorge Nunez, Xavier Otazu, Octavi Fors, Albert Prades, Vicenc Pala, Roman Arbiol, Multiresolution-based image fusion with additive wavelet decomposition, IEEE Trans. Geosci. Remote Sens. 37 (3) (1999) 1204–1211.
[34] Norden E Huang, Zheng Shen, Steven R Long, A new view of nonlinear water waves: the Hilbert spectrum, Annu. Rev. Fluid Mech. 31 (1) (1999) 417–457.
[35] Norden E Huang, Zheng Shen, Steven R Long, Manli C Wu, Hsing H Shih, Quanan Zheng, Nai-Chyuan Yen, Chi Chao Tung, Henry H Liu, The empirical mode decomposition and the Hilbert spectrum for nonlinear and non-stationary time series analysis, Proc. R. Soc. Lond. Ser. A Math. Phys. Eng. Sci. 454 (1971) (1998) 903–995.
[36] Thomas Kriechbaumer, Andrew Angus, David Parsons, Monica Rivas Casado, An improved wavelet–ARIMA approach for forecasting metal prices, Resour. Policy 39 (2014) 32–41.
[37] Chun-Fu Chen, Ming-Cheng Lai, Ching-Chiang Yeh, Forecasting tourism demand based on empirical mode decomposition and neural network, Knowl.-Based Syst. 26 (2012) 281–287.
[38] Shifei Zhou, Kin Keung Lai, Jerome Yen, A dynamic meta-learning rate-based model for gold market forecasting, Expert Syst. Appl. 39 (6) (2012) 6168–6173.
[39] Lu Xian, Kaijian He, Kin Keung Lai, Gold price analysis based on ensemble empirical model decomposition and independent component analysis, Physica A 454 (2016) 11–23.
[40] Fenghua Wen, Xin Yang, Xu Gong, Kin Keung Lai, Multi-scale volatility feature analysis and prediction of gold price, Int. J. Inf. Technol. Decis. Mak. 16 (01) (2017) 205–223.
[41] Konstantin Dragomiretskiy, Dominique Zosso, Variational mode decomposition, IEEE Trans. Signal Process. 62 (3) (2014) 531–544.
[42] Abhay Upadhyay, Ram Bilas Pachori, Instantaneous voiced/non-voiced detection in speech signals based on variational mode decomposition, J. Franklin Inst. B 352 (7) (2015) 2679–2707.
[43] Salim Lahmiri, Comparative study of ECG signal denoising by wavelet thresholding in empirical and variational mode decomposition domains, Healthc. Technol. Lett. 1 (3) (2014) 104–109.
[44] Yanxue Wang, Richard Markert, Jiawei Xiang, Weiguang Zheng, Research on variational mode decomposition and its application in detecting rub-impact fault of the rotor system, Mech. Syst. Signal Process. 60 (2015) 243–251.
[45] Salim Lahmiri, A variational mode decompoisition approach for analysis and forecasting of economic and financial time series, Expert Syst. Appl. 55 (2016) 268–273.
[46] Magnus R Hestenes, Multiplier and gradient methods, J. Optim. Theory Appl. 4 (5) (1969) 303–320.
[47] Danilo P Mandic, Jonathon Chambers, Recurrent Neural Networks for Prediction: Learning Algorithms, Architectures and Stability, John Wiley & Sons, Inc., 2001.
[48] Domingos S. de O. Santos Júnior, João F.L. de Oliveira, Paulo S.G. de Mattos Neto, An intelligent hybridization of ARIMA with machine learning models for time series forecasting, Knowl.-Based Syst. 175 (2019) 72–86.
[49] Zhan Bu, HuiJia Li, Jie Cao, Zhen Wang, Guangliang Gao, Dynamic cluster formation game for attributed graph clustering, IEEE Trans. Cybern. 49 (1) (2019) 328–341.
[50] HuiJia Li, Zhan Bu, Yulong Li, Zhongyuan Zhang, Yanchang Chu, Guijun Li, Jie Cao, Evolving the attribute flow for dynamical clustering in signed networks, Chaos Solitons Fractals 110 (2018) 20–27.

Please cite this article as: Y. Liu, C. Yang, K. Huang et al., Non-ferrous metals price forecasting based on variational mode decomposition and LSTM network, Knowledge-Based Systems (2019) 105006, https://doi.org/10.1016/j.knosys.2019.105006.